

Sociolinguistic Analysis of the Neomelodic Corpus

Exploratory Report on Lyrics

Research Team - Mafia Culture

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1 Introduction

This report documents the exploratory analysis of texts in the neomelodic corpus. The objective is not to estimate causal effects, but to reconstruct in a replicable way the distribution of themes, the distinctive lexicon associated with the main semantic clusters, and some descriptive heterogeneities with respect to time, artist birth area, gender, generation, track duration, text length, and sentiment.

The entire pipeline reported here was rerun in Stata. All the numbers discussed in the text derive from automatic project outputs, and the central tables are included through generated files in ‘analysis/lyrics/report/auto/’.

2 Dataset and Pipeline

2.1 Empirical Base

The working dataset includes 2601 tracks from the canonical high-confidence layer for which a usable lyric text is available at the title level. The unit of analysis is the track, not the artist. The performer’s demographic and geographic metadata are attached at the track level and are used only for the later descriptive stratifications.

2.2 Thematic Classification

Texts from the canonical layer are normalized to lowercase, stripped of punctuation, and classified through lexical dictionaries based on regular expressions. When available, the pipeline uses the Italian translation inherited from the old export; otherwise, it works directly on the song text. The final version of the dictionaries was made more conservative than in earlier iterations: the explicit goal is to privilege semantic precision over coverage.

The independent macro-themes are: Love, Family, Distance, Success, Money, Death, Territory, Faith, and Destiny. The **Malavita** macro-theme is defined as the aggregate of five sub-themes: Prison, Omertà, Respect, Fugitivity, and Police.

In the current sample, the share of tracks classified as **Malavita** is 25.3%.

2.3 Validation and Audit

For each track, the variable `main_themes` is built, concatenating the detected themes and making manual checking of matches easier. In addition, for the most fragile dictionaries

(polizia, territorio, destino, lontananza), a manual validation sample is exported to:

`analysis/lyrics/documentation/lyrics_regex_validation_samples.csv`

This report should therefore be read as a replicable exploratory audit of the pilot database, not as a definitive classification immune from residual ambiguities.

3 Thematic and Lexical Structure

3.1 Observed Dictionary Keywords

Rationale. This table checks whether the dictionaries are actually activating words observed in the corpus, rather than remaining abstract constructs that are not used by the texts.

Measure. Table 1 does not show “important” words in a substantive sense, but the dictionary patterns that effectively activate positive matches in the corpus. To avoid artificial tables, only keys with positive frequency are reported, and at most 5 items are shown per theme.

Interpretation. The correct reading is not “these are the most representative words of neomelodic music,” but “these are the operational keys that are materially triggering the classifier.”

Table 1: Theme-dictionary keywords with positive frequency observed in the corpus (N=2601 high-confidence songs). Counts indicate the number of texts in which at least one positive match of the pattern appears. At most 5 keys are shown for each theme.

Theme	Observed keywords (positive frequency)
<i>Independent Macro-Themes</i>	
Love	amore (1314), cuor (777), innamor (413), baci (195), core (174)
Family	mamm (334), figl (286), papà (190), frat (90), famigli (74)
Distance	aspett (483), manc (373), lontan (292), attes (22), assenz (14)
Success	success (100), ricch (30), fama (10), vittoria (6), popolar (4)
Money	sold (64), milion (29), guadagn (7), denar (6), miliard (6)
Death	mort (127), piant (74), bare (17), chiant (6), funeral (1)
Territory	napoli (123), palermo (18), sicilia (9), vesuvio (5), campania (2)
Faith	dio (446), preg (227), sant (98), chies (86), fede (29)
Destiny	strada (183), cammin (176), destin (88), fato (12), percor (12)
Malavita	aggregate (657)
<i>Malavita Macro-Category Sub-Themes</i>	
Prison	cella (144), prigion (43), carcer (17)
Omerta	silenzi (111), zitt (49), tace (16), nun parl (15), infame (9)
Respect	rispett (78), onor (50), boss (10)
Fugitivity	scapp (88), ombra (50), macchia (8), ricercat (3), latitant (2)
Police	divis (26), guardia (26), polizi (15), siren (11), carabinieri (3)

Stata code: `02_content_analysis_lyrics.do, lines 525–615. keep if count > 0; gsort section theme_order -count term_label; file open kwtext ...`

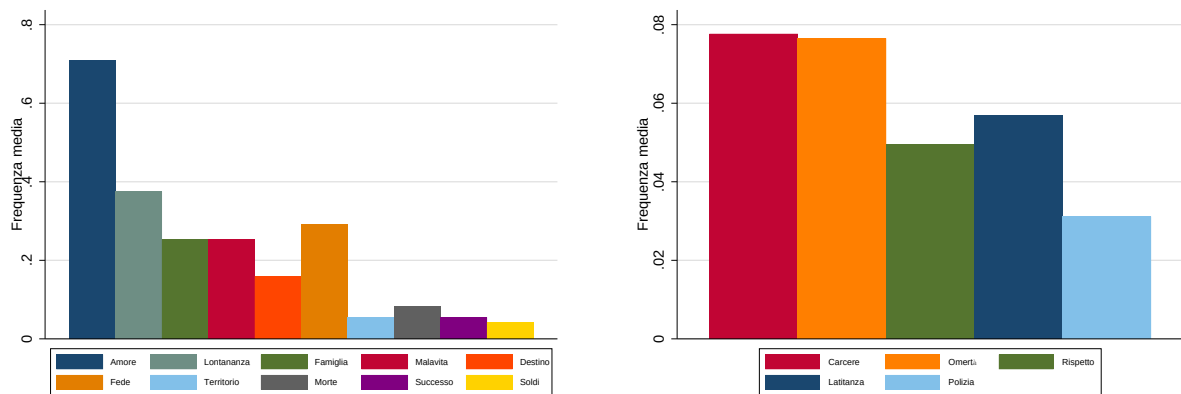
3.2 Theme Distribution and Correlations

Rationale. The goal here is to photograph the relative weight of themes and their elementary co-occurrence, without yet moving to inference.

Measure. The bars show the average shares of themes and malavita sub-themes; the matrices visualize correlations between binary indicators. On 0/1 variables, moderate correlations are normal and should not be read as absence of substantive relationship.

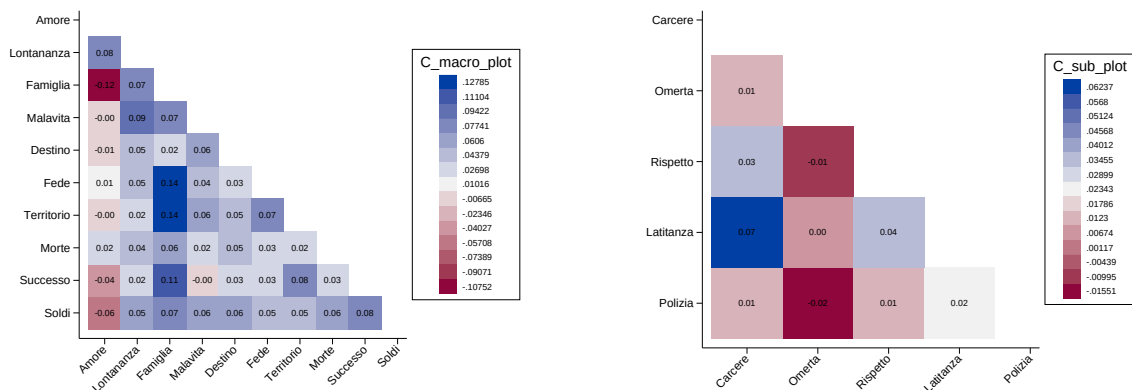
Interpretation. Aggregate frequencies confirm three simple facts. First, the sentimental sphere dominates the corpus. Second, distance and family remain highly visible axes. Third, malavita is not marginal: with a share equal to 25.3%, it is among the most present clusters in the textual corpus.

Figure 1: Descriptive frequencies of macro-themes and malavita sub-themes



Stata code: 03_cross_sectional_lyrics.do, lines 78-98. `graph bar (mean) ly_amore ... ly_soldi, asyvars; graph export ...`

Figure 2: Descriptive correlations among macro-themes and malavita sub-themes



Stata code: 03_cross_sectional_lyrics.do, lines 101-131. `quietly correlate ...; heatplot C_macro_plot ...; graph export ...`

3.3 Distinctive Lexicon and Word Clouds

Rationale. Word clouds are used to understand *which* lexicon distinguishes a theme, not just *how much* the theme is frequent.

Measure. To describe the distinctive vocabulary of themes, the lexical pipeline is entirely in Stata. The corpus is tokenized once, cleaned with Italian, Neapolitan, and technical stopwords, and then mapped back to aggregated lexical forms where appropriate. For the **general corpus**, the word cloud uses absolute frequencies. For specific themes, by contrast, a distinctiveness score relative to the complement is used:

$$\text{score}(w, T) = \left[\frac{f_T(w)}{\sum_v f_T(v)} \bigg/ \frac{f_{\bar{T}}(w) + 0.5}{\sum_v f_{\bar{T}}(v)} \right]^{0.45}$$

where $f_T(w)$ is the frequency of word w inside theme T and $f_{\bar{T}}(w)$ is its frequency in the complement. Beyond general stopwords, the most frequent dominant sentimental words are also excluded from non-love themes, to prevent romantic lexicon from obscuring secondary thematic markers.

Interpretation. The correct reading of the clouds is comparative: they highlight the *most distinctive* words, not simply the most frequent ones. In the case of *malavita*, for example, the current cloud primarily highlights the lexicon of prison, silence, escape, and control, not the simple high-frequency vocabulary of the corpus as a whole.

Figure 3: Lexical word clouds of the corpus and macro-themes



Notes: The general corpus uses absolute frequencies; specific themes use a distinctiveness score relative to the complement with 0.5 smoothing and exponent 0.45. **Stata code:** 02_content_analysis_lyrics.do, lines 363–409. `export_distinctive_word_counts "..."; shell python3 generate_wordcloud_lyrics.py`

4 Temporal and Demographic Heterogeneities

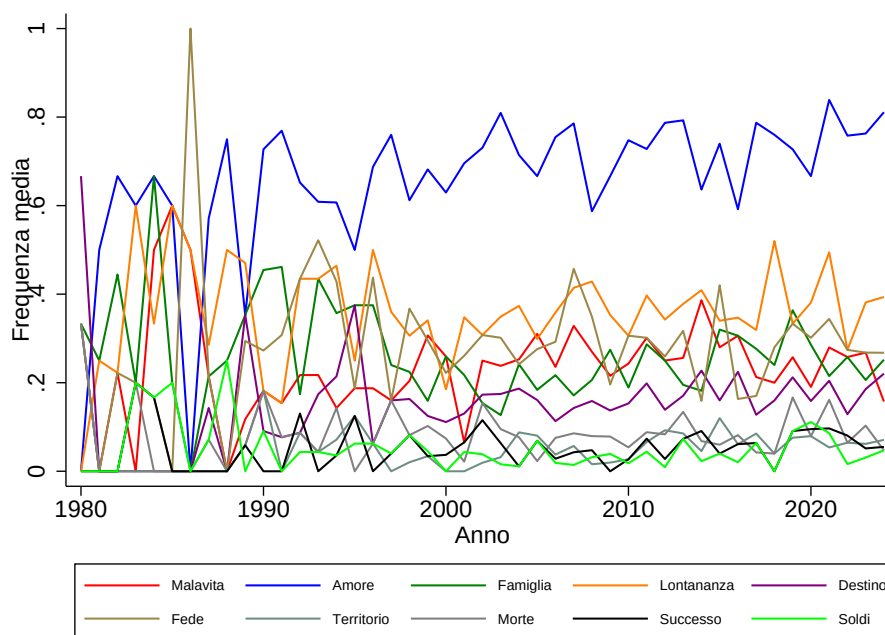
4.1 Temporal trends of macro-themes

Rationale. This analysis asks whether themes remain visible over time or instead appear only within restricted windows of the corpus.

Measure. The figure reports annual mean shares of macro-themes on the sub-sample with available lyrics.

Interpretation. Visually, the love theme remains the most present across the entire observed time span, while malavita stays at non-negligible and relatively stable levels. This section should be read as a description of the annual series, not as a test of a supposed historical break at an arbitrary threshold.

Figure 4: Temporal evolution of macro-themes



Notes: Annual mean shares computed on the sub-sample with available lyrics. The figure is descriptive. **Stata code:** 03_cross_sectional_lyrics.do, lines 41–61. `collapse (mean) ly_malavita ... by(release_year); twoway (line ...) ...`

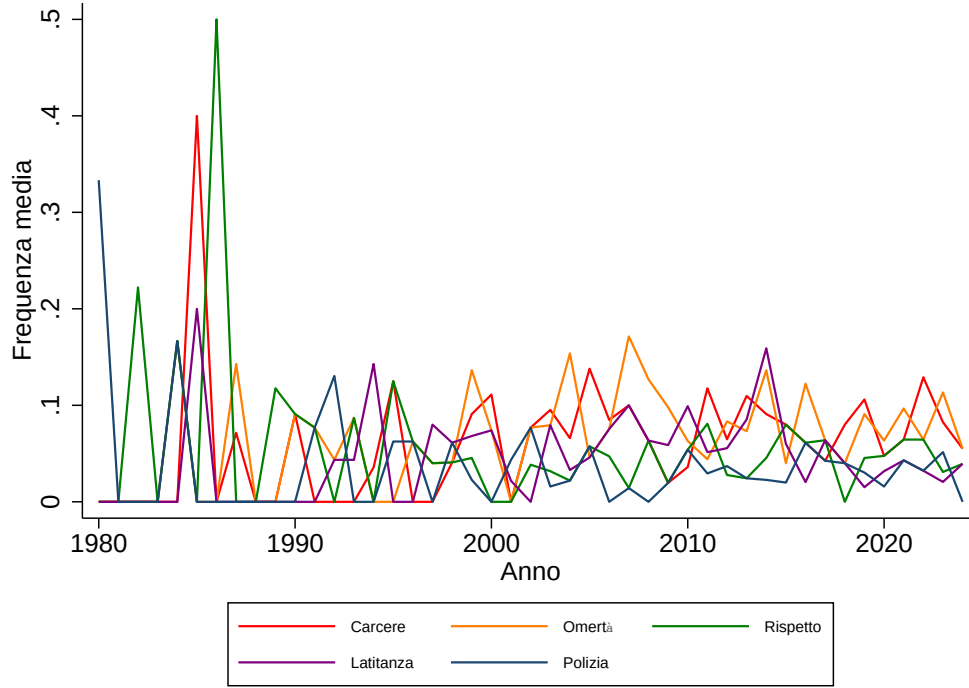
4.2 Trends of malavita sub-themes

Rationale. Once the malavita cluster is identified, it makes sense to decompose it in order to understand which narrative nuclei actually sustain it over time.

Measure. The figure shows the annual mean shares of the five sub-themes retained in the report.

Interpretation. The decomposition of the malavita cluster shows that the sub-themes do not all move in the same way. The prison component and the omertà/respect component remain visually central, whereas the less frequent themes should be read with greater caution.

Figure 5: Temporal evolution of malavita sub-themes



Notes: Descriptive series on the sub-themes of malavita. Rare themes should not be read as structural trends without further validation. **Stata code:** 03_cross_sectional_lyrics.do, lines 63–72. `twoway (line ly_carcere ...) ...; graph export ...`

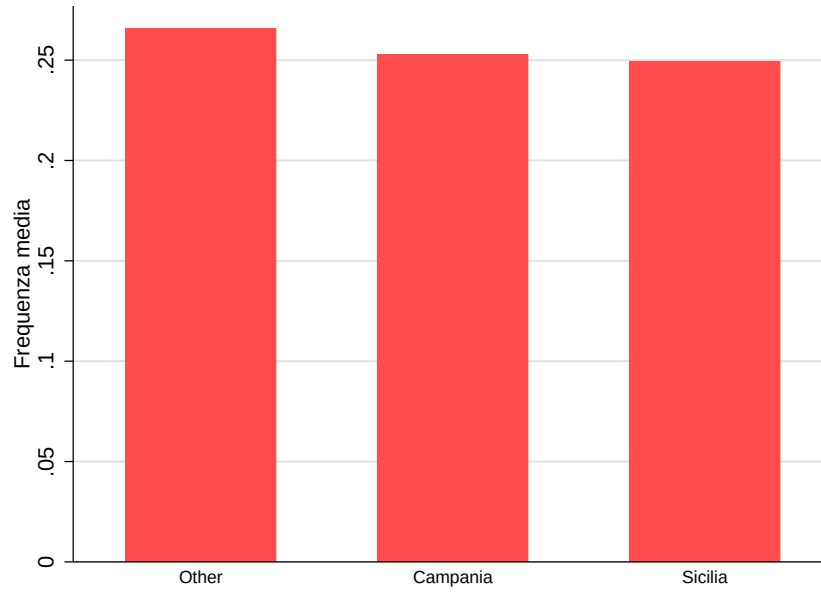
4.3 Geographic distribution

Rationale. This analysis checks whether the malavita theme remains confined to the areas historically more associated with the genre or is also spread elsewhere.

Measure. The mean malavita share is 25.3% in Campania (1962 tracks), 25.0% in Sicily (545 tracks), and 26.6% in other regions (94 tracks). Pairwise comparisons across groups always return high p-values: Campania vs Sicily $p = 0.877$, Campania vs Other $p = 0.775$, Sicily vs Other $p = 0.735$.

Interpretation. The correct reading is cautious: the criminal theme appears to be distributed fairly similarly across the three macro-groups, and with the available sample no geographically sharp statistical difference emerges.

Figure 6: Average prevalence of the malavita theme by birth area



Notes: Descriptive comparison among Campania, Sicily, and other regions. The “Other” group has a small sample size. **Stata code:** 03_cross_sectional_lyrics.do, lines 143–150. `collapse (mean) ly_malavita, by(region_group); graph bar ...`

4.4 Differences by gender

Rationale. The point here is to understand whether the thematic architecture changes systematically between women and men performers.

Measure. The following table reports the full set of comparisons for macro-themes and sub-themes, with means, Standard Errors, and t-test p-values.

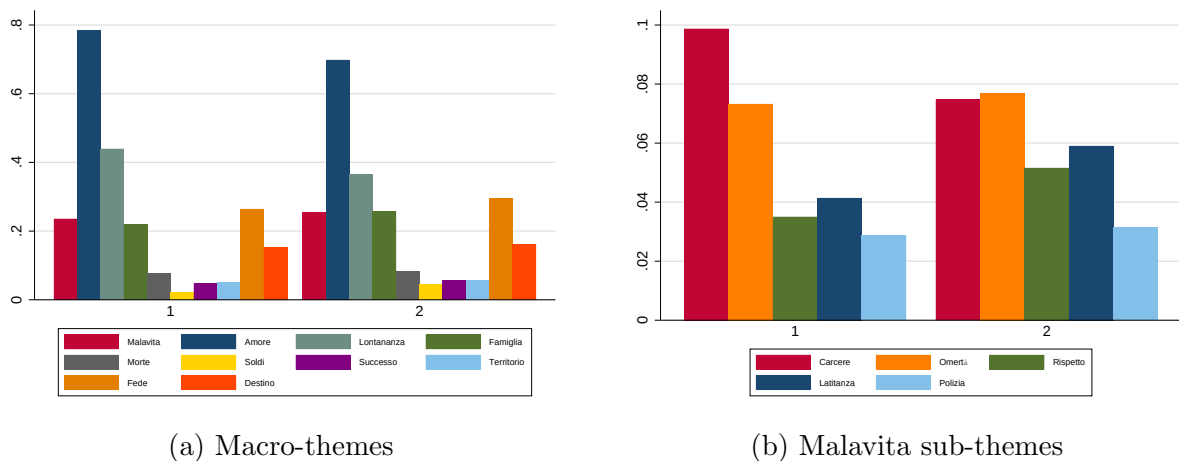
Interpretation. The aggregated malavita theme does not show appreciable differences between women and men: 23.6% against 25.5%, with $p = 0.462$. By contrast, some differences emerge on specific themes: love is more frequent among women performers (78.3% against 69.8%, $p = 0.002$), distance follows a similar direction (43.9% against 36.6%, $p = 0.012$), whereas prison is more present in women’s texts (9.9% against 7.5%, $p = 0.137$).

Table 2: Mean differences by gender for macro-themes and malavita sub-themes (sample N=2601 high-confidence lyrics). The table reports the mean, Standard Error (SE), Women-Men difference, and t-test p-value.

Theme	Women Mean (SE)	Men Mean (SE)	Diff.	p-value
<i>Macro-Themes</i>				
Love	78.3 (0.023)	69.8 (0.010)	8.5	0.002
Destiny	15.3 (0.020)	16.0 (0.008)	-0.8	0.730
Family	22.0 (0.023)	25.7 (0.009)	-3.7	0.153
Faith	26.4 (0.025)	29.6 (0.010)	-3.2	0.247
Distance	43.9 (0.028)	36.6 (0.010)	7.4	0.012
Malavita	23.6 (0.024)	25.5 (0.009)	-1.9	0.462
Death	7.6 (0.015)	8.3 (0.006)	-0.6	0.707
Money	2.2 (0.008)	4.5 (0.004)	-2.3	0.060
Success	4.8 (0.012)	5.6 (0.005)	-0.8	0.570
Territory	5.1 (0.012)	5.6 (0.005)	-0.5	0.739
<i>Malavita Sub-Themes</i>				
Prison	9.9 (0.017)	7.5 (0.006)	2.4	0.137
Fugitivity	4.1 (0.011)	5.9 (0.005)	-1.8	0.206
Omerta	7.3 (0.015)	7.7 (0.006)	-0.4	0.817
Police	2.9 (0.009)	3.1 (0.004)	-0.3	0.787
Respect	3.5 (0.010)	5.2 (0.005)	-1.7	0.205

Stata code: 03_cross_sectional_lyrics.do, lines 197–303. quietly ttest 'var_i', by(sex);
post gs ...; filefilter ... lyrics_gender_table.tex

Figure 7: Thematic distribution by artist gender



Notes: The differences are descriptive; pointwise tests are reported in the previous table. **Stata code:** 03_cross_sectional_lyrics.do, lines 306–324. graph bar (mean) ly_malavita ... over(sex);
graph export ...

4.5 Generational differences

Rationale. This section tests whether the thematic profile changes across the four cohorts that have readable sample sizes in the corpus.

Measure. For the four cohorts considered, the mean share of the malavita theme is 29.6% among Pre-Boomers (277 tracks), 23.2% among Baby Boomers (462), 26.2% among Gen X (1128), and 23.6% among Millennials (730). The following table reports, for each macro-theme and each malavita sub-theme, the means of the four main cohorts and the p-values of pairwise comparisons.

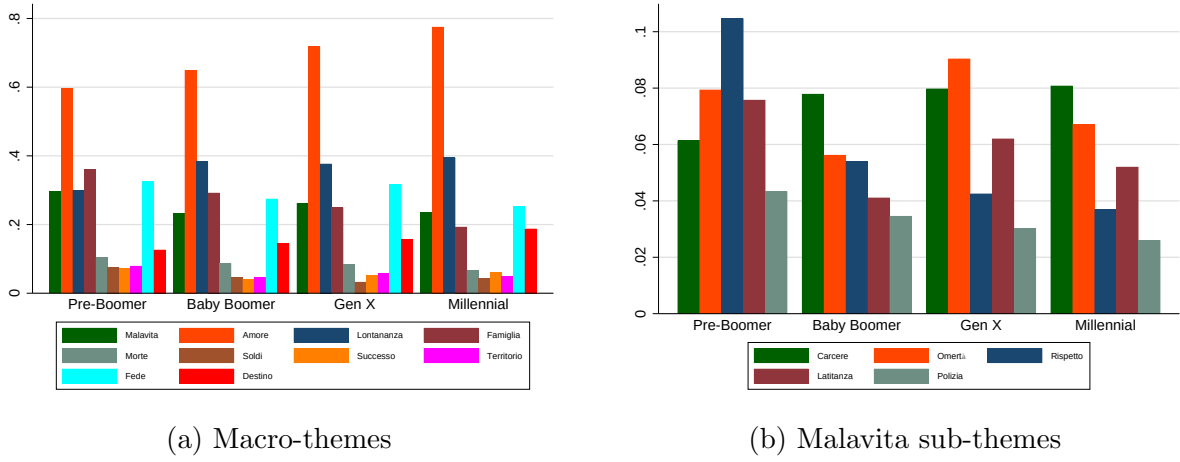
Interpretation. The tests should be read exploratorily: what matters here is mainly whether a systematic generational fracture emerges, not whether isolated differences can be ranked strongly. In the current corpus, this systematic fracture does not appear.

Table 3: Mean differences by generation for macro-themes and malavita sub-themes (Pre-Boomer, Baby Boomer, Gen X, Millennial only). The table reports cohort means and pairwise t-test p-values.

Theme	PreB	Boom	GenX	Mill	1-2	1-3	1-4	2-3	2-4	3-4
<i>Macro-Themes</i>										
Love	59.6	64.9	71.8	77.5	0.144	0.000	0.000	0.007	0.000	0.006
Destiny	12.6	14.5	15.6	18.8	0.477	0.216	0.021	0.580	0.057	0.075
Family	36.1	29.2	24.9	19.3	0.052	0.000	0.000	0.076	0.000	0.005
Faith	32.5	27.5	31.6	25.3	0.149	0.788	0.023	0.102	0.412	0.003
Distance	30.0	38.3	37.7	39.5	0.021	0.017	0.005	0.813	0.694	0.443
Malavita	29.6	23.2	26.2	23.6	0.052	0.259	0.049	0.200	0.873	0.194
Death	10.5	8.7	8.4	6.7	0.413	0.282	0.046	0.878	0.213	0.178
Money	7.6	4.5	3.3	4.2	0.085	0.001	0.033	0.222	0.806	0.279
Success	7.2	4.1	5.1	6.2	0.068	0.176	0.543	0.386	0.126	0.347
Territory	7.9	4.5	5.8	4.8	0.056	0.178	0.054	0.330	0.843	0.367
<i>Malavita Sub-Themes</i>										
Prison	6.1	7.8	8.0	8.1	0.399	0.301	0.297	0.901	0.857	0.936
Fugitivity	7.6	4.1	6.2	5.2	0.044	0.405	0.152	0.099	0.389	0.368
Omerta	7.9	5.6	9.0	6.7	0.217	0.563	0.496	0.023	0.453	0.073
Police	4.3	3.5	3.0	2.6	0.550	0.270	0.156	0.642	0.392	0.603
Respect	10.5	5.4	4.3	3.7	0.011	0.000	0.000	0.318	0.159	0.552

Stata code: 03_cross_sectional_lyrics.do, lines 400–520. `summarize 'var_i' if gen_cat == 1/4; ttest 'var_i' if inlist(gen_cat,...); filefilter ... lyrics_generation.table.tex`

Figure 8: Thematic distribution by generational cohort



Notes: The bars are descriptive; the previous table reports pairwise tests for all main cohorts. **Stata code:** 03.cross_sectional_lyrics.do, lines 388–398. `graph bar (mean) ly_malavita ... if inlist(gen_cat,1,2,3,4); graph export ...`

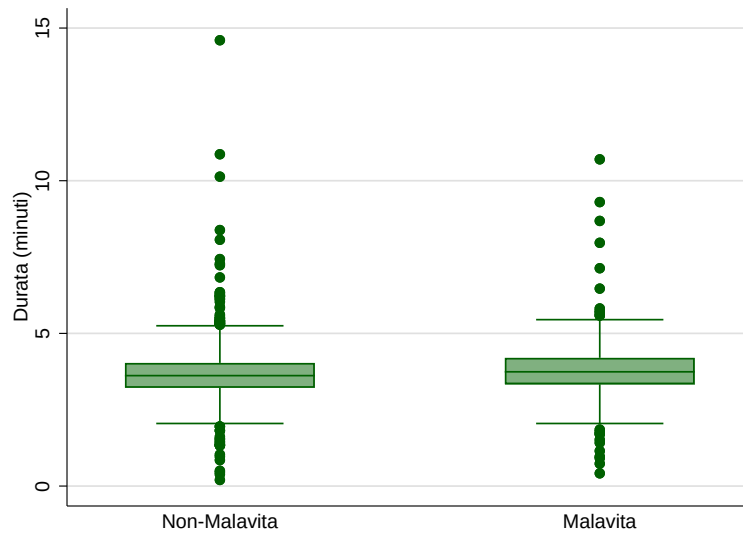
4.6 Track duration

Rationale. This check assesses whether malavita tracks are not only semantically different but also structurally somewhat longer in duration.

Measure. Tracks classified as malavita last on average 3.796 minutes, against 3.662 for the other texts; the t-test returns $p = 0.001$.

Interpretation. The difference is small in absolute value but statistically detectable in the sample. The result supports only one cautious conclusion: malavita tracks are *slightly* longer. On its own, this is not enough to demonstrate greater “narrative complexity.”

Figure 9: Track duration by presence of the malavita theme



Notes: Distribution of duration in minutes. **Stata code:** 03_cross_sectional_lyrics.do, lines 527–535. `ttest dur_min, by(ly_malavita); graph box dur_min, over(ly_malavita) ...`

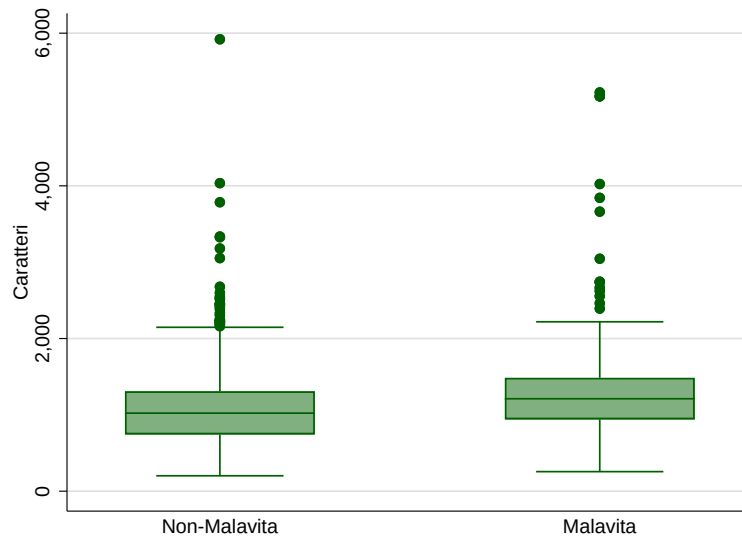
4.7 Text length

Rationale. Duration in minutes may fail to capture differences in textual density; for this reason, a separate measure of text extension is needed.

Measure. Mean text length is 1049.8 characters in non-malavita tracks ($SD = 455.5$) and 1252.0 in malavita tracks ($SD = 524.2$). The t-test returns $t = -9.46$ and $p = 0.000$.

Interpretation. Relative to duration, the differential is clearer here: malavita texts are on average longer. Again, however, the measure should be interpreted as an indicator of textual extension, not as direct proof of greater literary sophistication.

Figure 10: Text length by presence of the malavita theme



Notes: Number of characters in the translated text. **Stata code:** 03_cross_sectional_lyrics.do, lines 547–557. `ttest ly_length, by(ly_malavita); graph box ly_length, over(ly_malavita)` ...

5 Sentiment

Rationale. The sentiment section checks whether, beyond themes, the average lexical polarity of the texts also changes.

Measure. The sentiment section is now re-estimated directly on cleaned lyrics, without inheriting the score from the title pipeline. After tokenization and cleaning, matches to a conservative affective lexicon of positive and negative roots are counted for each track; the final score is defined as the lexical balance normalized by the number of text tokens, i.e. $(\text{positive} - \text{negative})/N_{\text{token}}$. Positive values indicate a relative prevalence of positive lexicon; negative values indicate a relative prevalence of negative lexicon.

Interpretation. In the current corpus, the sentiment lexicon identifies at least one positive or negative match in 61.5% of the tracks, with 1.67 average matches per text. In tracks without direct lexical hits from the dictionary, the score remains equal to zero: this is not missingness, but absence of matches. The overall mean score is -0.001, therefore very close to neutrality once normalized by length.

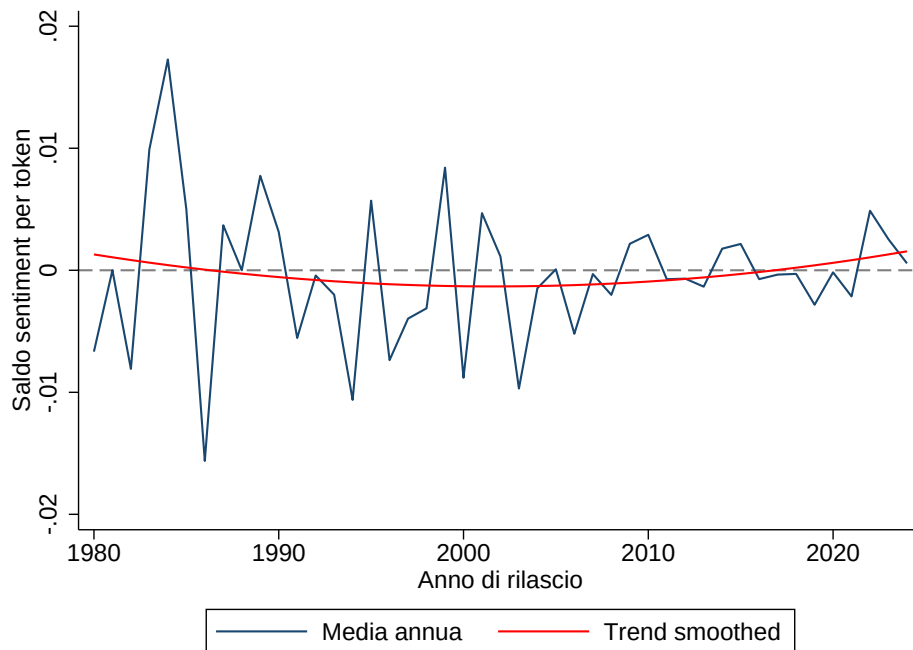
5.1 Temporal sentiment trend

Rationale. The idea here is to check whether the average polarity of the lexicon shifts over time.

Measure. The figure reports the average annual sentiment balance per token, accompanied by a purely descriptive smoothed line.

Interpretation. For sentiment as well, the correct reading is that of the overall annual trajectory, without imposing arbitrary breaks. The series oscillates around zero and does not suggest a clear secular drift toward positive or negative polarity.

Figure 11: Temporal trend of average sentiment



Notes: Descriptive series of the average annual sentiment balance per token, with smoothed line.
Stata code: 03_cross_sectional_lyrics.do, lines 562–573. `collapse (mean) sentiment_score, by(release_year); twoway (line ...) (lpoly ...) ...`

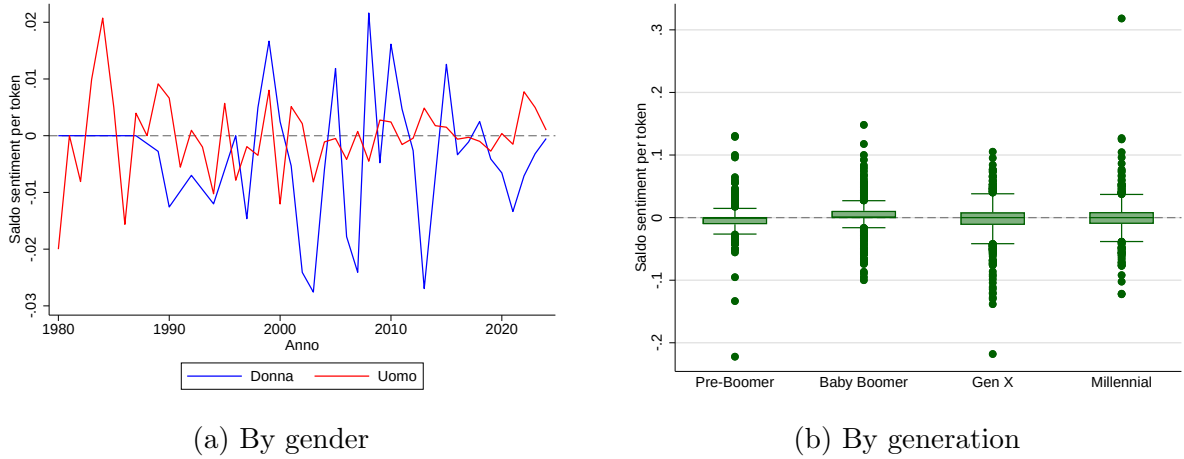
5.2 Sentiment by gender and generation

Rationale. This analysis checks whether average lexical tone changes with the performer's profile, not only with the thematic content of the track.

Measure. By gender, the means remain close: -0.004 among women and -0.000 among men, with $p = 0.014$. The comparison between Pre-Boomers and Millennials also does not signal a robust difference ($p = 0.813$).

Interpretation. In the available corpus, no strong polarization of sentiment emerges by artist demographic characteristics.

Figure 12: Sentiment by artist characteristics



Notes: The figures should be read as descriptive distributions of the sentiment score. The generational boxplot includes only Pre-Boomer, Baby Boomer, Gen X, and Millennial. **Stata code:** `03_cross_sectional_lyrics.do`, lines 594–613. `collapse (mean) sentiment_score, by(release_year sex); graph export sentiment_by_sex.pdf; graph box sentiment_score if inlist(gen_cat,1,2,3,4) ...`

5.3 Sentiment and themes

Rationale. The last step connects thematic structure and average emotional polarity, to see whether some themes carry a darker or more positive lexical balance.

Measure. Non-malavita tracks have a mean score equal to 0.000, against -0.004 for malavita tracks; the t-test returns $p = 0.000$. The theme panels remain useful descriptive support for seeing how this balance is distributed across the main macro-themes and the five retained malavita sub-themes.

Interpretation. The difference is statistically visible and moves in the expected direction: the lexicon of malavita tracks is, on average, more negative. The absolute size of the gap remains limited, however, so the correct reading is that of an average shift in lexical balance, not a sharp opposition between two completely separate emotional universes.

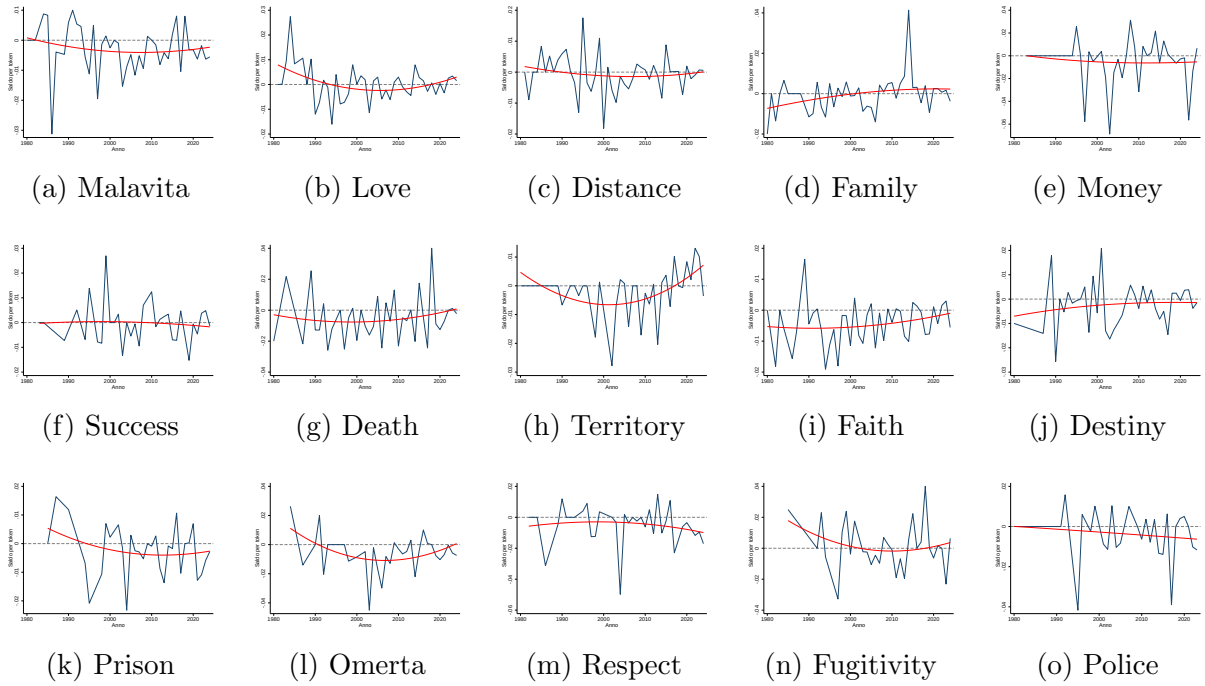
6 Conclusions

The current pipeline delivers an exploratory report that is finally traceable. The main results are four.

First, the sentimental sphere remains dominant, but malavita occupies a substantial share of the textual corpus (25.3%) and does not appear as a residual phenomenon. Second, the presence of criminal lexicon remains visible across the whole observed time span, without requiring this report to force arbitrary chronological breaks. Third, the aggregated malavita theme does not show strong differences by gender or geographic area in the available comparisons, although more specific heterogeneities do emerge for some sub-themes and on the love/distance side. Fourth, malavita tracks are on average somewhat longer and, above all, have longer texts; once sentiment is correctly re-estimated on lyrics, they also show a more negative average lexical balance, though of moderate magnitude.

On the scientific plane, this report does not settle the question of the semantic validity of the dictionaries, but it does raise their transparency considerably: classification, lexical cleaning, exports, word clouds, and central tables are now regenerable and readable as a coherent audit of the pilot database.

Figure 13: Temporal sentiment panels for macro-themes and malavita sub-themes



Notes: Each panel shows the average annual sentiment balance for the sub-sample in which the theme is present. Less dense panels should be read as descriptive support, not as stable series. **Stata code:** 03_cross_sectional_lyrics.do, lines 577–614. `foreach tema in malavita amore ... polizia; keep if ly_tema == 1; collapse (mean) sentiment_score, by(release_year); twoway ...`